I want to give a slight commentary of

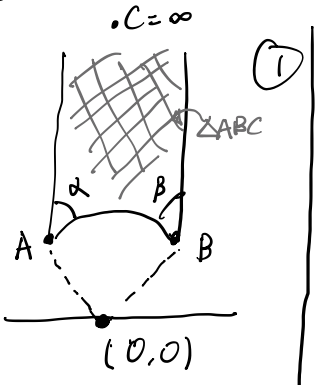
Theorem 5.24. A hyperbolic triangle with angle α, β, γ has area $\pi - \alpha - \beta - \gamma$.

Main Ingredients: [For the general case, see lecture.]

↳ Hyperbolic Isometries ([Compositions of] Inversion of unit circle, Dilations, Horizontal Translations, Vertical Reflections) preserves Areas, and Angles — Prop. 4.14

↳ Specifying [Hyperbolic Line → Hyperbolic Line, Point ↦ Point] determines an hyp. isometry. — Prop. 5.20

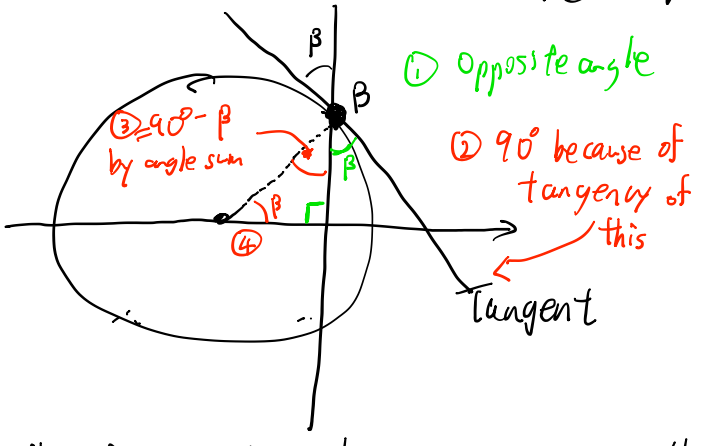
Proof. Let us look at a special case first.

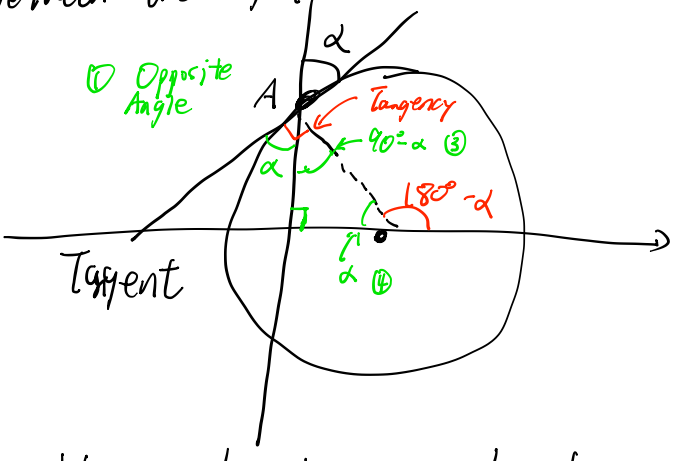


① [AB is a hyperbolic line (arc through unit circle)
C = ∞ so AC, BC are vertical line.]

Question: [In Café Session] What is the coordinate of A and B? Answer: It is $\begin{cases} A = (\cos(\pi-\alpha), \sin(\pi-\alpha)) \\ B = (\cos\beta, \sin\beta) \end{cases}$

Why? Remember: the angle between two lines is the angle between the tangents:





And the fact at angle $0 < \theta < 2\pi$ the unit circle has coordinate $(\cos\theta, \sin\theta)$



finishes our claim (i.e. $A = (\cos(\pi-\alpha), \sin(\pi-\alpha))$, $B = (\cos\beta, \sin\beta)$)

So $\triangle ABC$ is parametrized by $\{(x,y) \mid \cos(\pi-\alpha) < x < \cos(\beta)$ and $\sqrt{1-x^2} < y < \infty\}$.

($x^2+y^2>1$ ⇒ Region outside unit circle; The horizontal strip)

$\therefore$ Area $(\triangle ABC) = \int_{\cos(\pi-\alpha)}^{\cos(\beta)} \int_{\sqrt{1-x^2}}^{\infty} \frac{1}{y^2}\, dx\, dy = \pi - \alpha - \beta.$

($\cdots \{(EG-F^2)^{1/2} \quad E = \frac{1}{y^2},\ F = 0,\ G = \frac{1}{y^2}$)

follow notes =